

A Withheld Timing Test of 3XMM J215022.4–055108

Independent X-ray Spectral Closure and CPTG Compact-State Readout

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Abstract

3XMM J215022.4–055108 is an off-nuclear X-ray transient and one of the strongest intermediate-mass black-hole candidates, with its long-lived thermal flare interpreted as the tidal disruption of a star in a massive star cluster. Its later discovery of a coherent quasi-periodic oscillation near 85 s provides an unusually sharp timing constraint on a mass range that remains difficult to probe directly. We construct a retrospective withheld-input test by using a fully relativistic multi-epoch spectral-energy-distribution analysis that excludes the XMM-Newton observation containing the oscillation. That independent analysis reports an intrinsic disk inner radius $\log_{10}(R_{\text{in}}/\text{km}) = 5.24^{+0.13}_{-0.05}$ and a derived mass $\log_{10}(M_{\bullet}/M_{\odot}) = 4.39^{+0.17}_{-0.08}$. The observed 11.88 ± 0.38 mHz signal corresponds to 12.5334 ± 0.4009 mHz in the source frame. Curvature Polarization Transport Gravity (CPTG) retains the Kerr exterior while assigning a finite compact interior and a fixed source–geometry mass–spin decomposition. Conditional on identifying the oscillation with Lense–Thirring precession at the ISCO, the independently fitted radius and withheld timing signal close at exterior spin $q = 0.5005$ and exterior gravitational mass $M_{\infty} = 2.771 \times 10^4 M_{\odot}$, inside the separately reported spectral mass support. The registered CPTG compact relation then gives $M_{\text{c}} = 2.031 \times 10^4 M_{\odot}$ and $M_{\text{geom}} = 7.397 \times 10^3 M_{\odot}$, a 26.7% geometric mass-equivalent share, without object-specific compact retuning. The result is a withheld exterior-timing closure followed by a finite CPTG state readout; it does not identify the QPO itself with an internal CPTG clock.

1 Introduction

Black holes are established observationally at stellar masses and at the supermassive scales found in galactic nuclei, but the intermediate regime remains much harder to populate securely. Intermediate-mass black holes (IMBHs), broadly in the 10^3 – $10^5 M_\odot$ range relevant here, are scientifically important because they occupy the mass gap between ordinary stellar remnants and supermassive black holes and may preserve information about how massive black-hole seeds form, grow, and survive in dense stellar systems. Their relative faintness and scarcity make direct dynamical measurements difficult, so transient accretion events provide an unusually valuable route to finding and characterizing them [1].

3XMM J215022.4–055108 (hereafter 3XMM J2150) is especially useful because it is not a conventional persistent galactic nucleus. It is an off-nuclear X-ray transient associated with a massive stellar system at redshift $z = 0.055$. Multiwavelength follow-up found a marginally resolved optical counterpart consistent with a star cluster and an X-ray luminosity and supersoft thermal evolution consistent with a stellar tidal-disruption event (TDE), removing an earlier Galactic cooling-neutron-star interpretation and establishing the source as one of the strongest IMBH candidates [3, 4, 1]. In physical terms, the disruption temporarily illuminates an otherwise difficult-to-measure compact object by creating a hot accretion flow whose spectrum carries information about the characteristic inner scale.

The source became more unusual when a coherent X-ray quasi-periodicity was identified in a later XMM-Newton observation. The measured centroid is

$$\nu_{\text{obs}} = 11.88 \pm 0.38 \text{ mHz}, \tag{1}$$

corresponding to an observed Fourier period near 84 s and an epoch-folded period near 85 s. The feature has quality factor $Q \sim 16$, fractional rms amplitude of order 10%, and a reported conservative significance above 3.5σ before additional independent-search accounting [1]. If the signal is tied to a relativistic orbital or precessional timescale, it acts as a clock set by black-hole mass and spin. That possibility makes 3XMM J2150 more than an IMBH candidate: it becomes a rare laboratory in which an independently inferred disk scale can be confronted with a strong-field timing observable.

The most important feature for the present test is that the spectral and timing information need not be taken from the same observation. Guolo et al. later modeled 3XMM J2150 with a fully relativistic multi-wavelength disk calculation using an earlier XMM-Newton spectrum and a much later Chandra epoch, neither of which is the XMM2 observation containing the reported QPO [2]. This separation permits a retrospective withheld-input test. The disk scale can be fixed first; only afterward is the 85-s timing signal revealed and used to ask whether the two measurements select a mutually consistent exterior mass–spin state. Here, “withheld” refers to the information flow of the present analysis rather than to a prospectively preregistered prediction.

Curvature Polarization Transport Gravity (CPTG) is a geometric extension of General Relativity in which ordinary physical stress–energy remains the material source of gravity

while two additional geometric response sectors describe how curvature is organized. Curvature polarization describes the reorganization of the sourced gravitational field, and curvature transport describes how that organized response is carried through an extended or evolving geometry [7]. General Relativity is recovered continuously wherever these additional responses become negligible. CPTG therefore does not replace the successful Kerr description of an exterior black-hole spacetime merely to fit this QPO.

The compact-object realization adds a different requirement in the deeper strong-field region: the completed physical interior remains finite and joins the ordinary Kerr or Schwarzschild exterior consistently [9]. A registered mass–spin relation—meaning a relation fixed in the published compact formulation before 3XMM J2150 is analyzed—then separates the localized physical compact-source mass from the geometric contribution to the conserved exterior gravitational charge. Once that compact state is solved, Structural Mode N provides a post-solution measurement of how the finite geometry is internally organized; it is not a fitting parameter and is not used to choose the observed timing frequency [10]. These capabilities make the theory testable in a layered way. An exterior observation first fixes the gravitational state in the regime where CPTG and Kerr overlap; CPTG then makes additional statements about the finite source–geometry state behind that same exterior field.

This paper applies that logic to 3XMM J2150. The independent spectral inner radius is used as the primary construction datum, the QPO is withheld until the timing stage, and the separately reported spectral mass is retained as a cross-check rather than reused as an independent input from the same posterior. The exterior state is solved under the specific hypothesis that the QPO is Lense–Thirring precession at the innermost stable circular orbit (ISCO). Only after that state closes is the registered CPTG compact map evaluated. Because the problem combines an external relativistic disk model, observer-to-source timing, a first-order comparison-coordinate sensitivity translation, Kerr timing, uncertainty reconstruction, and a finite CPTG readout, the complete procedure is stated explicitly in the following Methodology section before any numerical result is presented.

The resulting claim is specific. A successful result establishes that an independently constructed disk state survives a withheld timing constraint under the tested LT-at-ISCO hypothesis and that the resulting exterior state passes through the fixed CPTG compact map without object-specific retuning. The 85-s signal remains an exterior timing constraint in this construction; the finite-interior quantities are downstream CPTG readouts of the closed exterior state.

2 Methodology

The analysis is deliberately layered so that quantities inferred under one physical model are not silently reused as though they were direct observables of another. First, the QPO-detection epoch is separated from the X-ray data used to construct the independent spectral-energy-distribution state. Second, the observed timing frequency is translated to the source-frame asymptotic timing coordinate using the measured redshift. Third, the published physical disk scale is exposed as a comparison coordinate and its small CPTG luminosity-coordinate sensitivity is evaluated explicitly. Fourth, the retained Kerr exterior

is used to solve the conditional LT-at-ISCO mass–spin problem. Fifth, the separately reported spectral mass is used only as a consistency check. Finally, and only after the exterior state has closed, the registered CPTG compact mass–spin and Structural-Mode operators are evaluated. No CPTG internal quantity is allowed to flow backward into the timing solution.

2.1 The withheld XMM2 QPO

Zhang et al. searched the available XMM-Newton light curves of 3XMM J2150 and found the QPO in a ~ 17 ks good-time segment of the second XMM-Newton observation [1]. The measured peak is Eq. (1); the authors report an observed period of about 84 s from the Fourier peak and 85 s from epoch folding. Their timing analysis gives $Q \sim 16$ and a folded fractional rms amplitude of $10.1 \pm 2.3\%$.

The QPO is therefore withheld from construction of the primary spectral radius and enters only after that independent spectral quantity has been fixed. This preserves the observational separation stated in the Introduction while keeping the timing datum available as the subsequent exterior-state test.

Zhang et al. evaluate Keplerian, vertical epicyclic, and LT precession frequencies at the ISCO. Their same-epoch XMM2 continuum contours overlap the timing curves only under the LT identification, yielding approximately

$$9.9 \times 10^3 \lesssim M_{\text{BH}}/M_{\odot} \lesssim 1.6 \times 10^4, \quad 0.26 \lesssim q \lesssim 0.36 \quad (2)$$

at their quoted joint 90% level [1]. Those same-epoch values are retained in this paper as an external comparison, not as inputs to the withheld test.

2.2 Independent relativistic SED constraint

Guolo et al. fit TDE spectral energy distributions with the fully relativistic **kerrSED** model, whose intrinsic parameters include R_{in} , black-hole spin, and inclination [2]. The inner boundary is identified with the ISCO, while relativistic photon propagation is evaluated by numerical ray tracing. For 3XMM J2150, the relevant observations are:

Epoch	Approx. time	X-ray observation	Role in the withheld test
E1	179 ± 1 d	XMM-Newton/PN 0404190101	Independent spectral epoch
E2	3940 ± 630 d	Chandra/ACIS 17862	Independent late spectral epoch
QPO epoch	Separate	XMM-Newton/PN 0603590101 (GTI2)	Withheld timing observation

Table 1: Observational separation used by the withheld test. The first two entries are the 3XMM J2150 epochs used in the multi-epoch SED analysis of Guolo et al.; the QPO is reported from GTI2 of the distinct XMM2 observation 0603590101 by Zhang et al.

For 3XMM J2150 the SED fit reports

$$\log_{10}\left(\frac{R_{\text{in}}}{\text{km}}\right) = 5.24^{+0.13}_{-0.05}, \quad (3)$$

and

$$\log_{10}\left(\frac{M_{\bullet}}{M_{\odot}}\right) = 4.39^{+0.17}_{-0.08}. \quad (4)$$

The distinction between these quantities matters. In the `kerrSED` inference, R_{in} is an intrinsic free parameter, while $M_{\bullet}$ is a secondary quantity calculated element-by-element from the posterior [2]. We therefore use Eq. (3) as the primary spectral input and Eq. (4) as a withheld cross-check on the mass derived from timing closure.

Numerically, Eq. (3) corresponds to

$$R_{\text{in},50}^{\text{pub}} = 1.73780 \times 10^5 \text{ km}, \quad (5)$$

with published 68% limits

$$1.54882 \times 10^5 < R_{\text{in}}^{\text{pub}}/\text{km} < 2.34423 \times 10^5. \quad (6)$$

The spectral radius remains a model-derived comparison coordinate, not a raw observable. It inherits the disk assumptions and distance convention of the external fit. This paper therefore tests consistency with that published spectral construction rather than relabeling it as a CPTG-native measurement.

2.3 Observed and source-frame timing coordinates

The reported QPO frequency is measured in the observer frame. The Kerr timing operator is defined with respect to asymptotic coordinate time in the source rest frame, so the direct spectroscopic redshift is used to restore the corresponding source-frame frequency:

$$\nu_{\text{src}} = (1 + z)\nu_{\text{obs}}. \quad (7)$$

For $z = 0.055$,

$$\nu_{\text{src}} = 12.5334 \pm 0.4009 \text{ mHz}. \quad (8)$$

The center of this frequency interval corresponds to a source-frame period of 79.79 s.

This is a time-coordinate correction, not an insertion of a Hubble constant into the compact calculation. The measured redshift is used directly. For transparency, we also retain an observed-frequency benchmark in the numerical results. Using the published 11.88 mHz value without the factor $(1 + z)$ reproduces the published timing convention closely; the source-frame calculation of Eq. (8) is the primary compact-state calculation in this paper.

2.4 Spectral comparison-coordinate translation

The Guolo et al. fit adopts a standard Λ CDM comparison cosmology with

$$H_{0,\text{ref}} = 73 \text{ km s}^{-1} \text{ Mpc}^{-1} \quad (9)$$

[2]. The paper does not state the remaining density parameters explicitly, so an exact reconstruction of its luminosity-distance convention from the manuscript alone is underdetermined. We therefore do not silently promote the reported physical radius to a CPTG-native coordinate.

For an explicit low-redshift sensitivity check, we decode the published scale with a flat comparison model

$$\Omega_{m,\text{ref}} = 0.3, \quad \Omega_{\Lambda,\text{ref}} = 0.7, \quad (10)$$

while the registered CPTG affine luminosity comparison coordinate and geometric π -branch expansion law are taken from the CPTG comparison-coordinate authority [8] and use

$$H_{0,L}^{\text{aff}} = 73.1711530416 \text{ km s}^{-1} \text{ Mpc}^{-1}, \quad (11)$$

and

$$E_{\pi}(z)^2 = \left(1 - \frac{1}{\pi}\right) + \frac{1}{\pi}(1+z)^{\pi}. \quad (12)$$

At $z = 0.055$, the two luminosity-distance constructions give

$$D_{L,\text{ref}} = 235.3122 \text{ Mpc}, \quad D_{L,\text{CPTG}} = 234.4321 \text{ Mpc}. \quad (13)$$

Their ratio is

$$\mathcal{R}_D = \frac{D_{L,\text{CPTG}}}{D_{L,\text{ref}}} = 0.9962597. \quad (14)$$

For a fixed relativistic disk spectral shape, the inferred linear disk scale follows the distance normalization to first order. We therefore apply the ratio only as a first-order comparison-coordinate sensitivity rescaling and use

$$R_{\text{in}}^{\text{CPTG}} = \mathcal{R}_D R_{\text{in}}^{\text{pub}} = 1.73130 \times 10^5 \text{ km} \quad (15)$$

as the primary first-order CPTG comparison-coordinate sensitivity value. The superscript labels this sensitivity translation; it does not denote a new CPTG refit of the SED. Applying the same scale factor to the published 68% radius limits gives

$$1.54302 \times 10^5 < R_{\text{in}}^{\text{CPTG}} / \text{km} < 2.33546 \times 10^5. \quad (16)$$

This is only a 0.374% shift from the published radius, far below the reported radius uncertainty. All principal conclusions are also checked in the unshifted published coordinate. The operation is a first-order sensitivity translation of the reported physical scale, not a reanalysis of the original SED data; its purpose is to expose the external coordinate convention rather than to obtain closure by cosmological retuning.

2.5 CPTG compact-state framework

The exterior timing solution supplies the conserved gravitational mass and spin that are visible outside the compact object. CPTG then maps that already closed exterior state into a finite source–geometry decomposition and post-solution structural readouts. This subsection states those operators in the order in which they are evaluated; none is used to tune the preceding spectral or timing data.

2.5.1 Exterior recovery and finite interior

The parent CPTG field organization is

$$G_{\mu\nu} + \Pi_{\mu\nu}^{\text{pol}} + \Theta_{\mu\nu}^{\text{trans}} = \frac{8\pi G}{c^4} T_{\mu\nu}, \quad (17)$$

where ordinary physical stress–energy remains the material source, while curvature polarization and curvature transport are geometric response sectors [7]. In the registered compact realization, the deeper physical continuation remains finite and joins an ordinary Kerr or Schwarzschild exterior where the additional response is negligible [9].

The compact matching scale is

$$r_m = \frac{GM_\infty}{c^2}, \quad (18)$$

with M_∞ the conserved exterior gravitational mass parameter. For

$$q = |\hat{a}_\infty|, \quad (19)$$

the retained Kerr outer horizon is

$$r_+^{\text{Kerr}} = r_m \left(1 + \sqrt{1 - q^2} \right). \quad (20)$$

The handoff radius is not a universal polarization–transport activation switch. It is the matching boundary of the registered finite realization; the detailed crossover remains object dependent [9].

2.5.2 Conditional mass–spin closure

The dimensionless factor A_{pol} is the registered compact polarization factor that connects the conserved exterior gravitational state to the localized compact-source contribution while preserving the mass–spin handoff. It is fixed by the exterior spin through the registered compact relation

$$q^2 A_{\text{pol}}^3 + A_{\text{pol}} - 2 = 0, \quad (21)$$

with the physical branch

$$1 \leq A_{\text{pol}} \leq 2. \quad (22)$$

After Eq. (21) closes,

$$M_c = \frac{M_\infty}{A_{\text{pol}}}, \quad M_{\text{geom}} = M_\infty - M_c, \quad (23)$$

and angular-momentum conservation gives

$$\hat{a}_c = A_{\text{pol}}^2 \hat{a}_\infty. \quad (24)$$

No 3XMM-specific compact coefficient is introduced. The QPO, disk radius, or disk temperature does not modify Eq. (21).

2.5.3 Post-solution Structural Mode

The finite compact profile is

$$F(x) = \frac{1}{8} (63x^5 - 90x^7 + 35x^9), \quad x = \frac{r}{r_m}, \quad (25)$$

with dimensionless acceleration shape

$$U(x) = \frac{F(x)}{x^2}. \quad (26)$$

The compact structural invariant and weight are

$$\mathcal{C}(x) = U(x)^2 + U'(x)^2 + 2 \left[\frac{U(x)}{x} \right]^2, \quad w(x) = |\mathcal{C}'(x)|. \quad (27)$$

For the rotating equal-measure coordinate

$$\xi(x, q) = \left[\frac{x(x^2 + q^2)}{1 + q^2} \right]^{1/3}, \quad (28)$$

the compact Structural Mode is

$$N_{\text{BH}}(q) = \frac{\int_0^1 w(x) dx}{\int_0^1 \xi(x, q) w(x) dx}. \quad (29)$$

The corresponding organization shell follows from

$$x_N(x_N^2 + q^2) = \frac{1 + q^2}{N_{\text{BH}}^3}, \quad r_N = x_N r_m. \quad (30)$$

Structural Mode is evaluated only after the compact state has closed and therefore cannot influence the QPO frequency, mass, or spin [10].

2.5.4 Trapping, clock, and frame-dragging readouts

The remaining compact quantities reported later are downstream geometric readouts of the same solved state. Using the finite profile of Eq. (25), define

$$\delta(x, q) = x^2 - 2xF(x) + q^2, \quad (31)$$

$$\sigma(x, \theta, q) = x^2 + q^2 \cos^2 \theta, \quad (32)$$

$$\mathcal{A}_d(x, \theta, q) = (x^2 + q^2)^2 - q^2 \delta(x, q) \sin^2 \theta. \quad (33)$$

For the subextremal branch used here, the inner trapping surface is the root $0 < x_- < 1$ of

$$\delta(x_-, q) = 0, \quad r_-^{\text{CPTG}} = x_- r_m. \quad (34)$$

For a zero-angular-momentum observer (ZAMO), the local lapse and its determinant-measure angular average are

$$\alpha(x, \theta; q) = \sqrt{\frac{\delta \sigma}{\mathcal{A}_d}}, \quad (35)$$

$$\bar{\alpha}(x; q) = \frac{\int_0^\pi \alpha(x, \theta; q) \sigma(x, \theta, q) \sin \theta d\theta}{\int_0^\pi \sigma(x, \theta, q) \sin \theta d\theta}. \quad (36)$$

The Structural- N clock is then read at the organization shell,

$$T_{24,N}(q) = 24 \bar{\alpha}(x_N(q); q) \text{ h}, \quad (37)$$

which measures proper-time accumulation there during 24 source-frame asymptotic exterior Kerr hours. The corresponding geometric ZAMO frame-dragging field and its angular average are

$$\omega(x, \theta; q) = \frac{c}{r_m} \frac{2q x F(x)}{\mathcal{A}_d(x, \theta, q)}, \quad (38)$$

$$\bar{\omega}(x; q) = \frac{\int_0^\pi \omega(x, \theta; q) \sigma(x, \theta, q) \sin \theta d\theta}{\int_0^\pi \sigma(x, \theta, q) \sin \theta d\theta}, \quad (39)$$

with $\bar{\omega}_N \equiv \bar{\omega}(x_N; q)$. These clock, trapping, and frame-dragging quantities are defined by the registered compact realization and are evaluated only after (M_∞, q) has closed [9, 10].

2.6 Withheld-timing construction

The observational closure is performed entirely in the retained Kerr exterior. The independent disk scale and the source-frame QPO frequency determine the exterior mass–

spin state under the stated LT-at-ISCO identification; only afterward is that state passed into the CPTG compact operators above.

2.6.1 Kerr timing law retained by CPTG

For prograde circular motion in the retained Kerr exterior, let $\rho \equiv R/r_m$ denote the dimensionless Boyer–Lindquist radius. The dimensionless azimuthal orbital frequency is

$$\Omega_\phi = \frac{1}{\rho^{3/2} + q}, \quad (40)$$

and vertical epicyclic frequency

$$\Omega_\theta = \Omega_\phi \sqrt{1 - \frac{4q}{\rho^{3/2}} + \frac{3q^2}{\rho^2}}. \quad (41)$$

The physical nodal-precession frequency is

$$\nu_{\text{LT}} = \frac{c^3}{2\pi G M_\infty} (\Omega_\phi - \Omega_\theta). \quad (42)$$

The dimensionless ISCO radius is written as

$$Z_1 = 1 + (1 - q^2)^{1/3} \left[(1 + q)^{1/3} + (1 - q)^{1/3} \right], \quad (43)$$

$$Z_2 = \sqrt{3q^2 + Z_1^2}, \quad (44)$$

$$\rho_{\text{ISCO}}(q) = 3 + Z_2 - \sqrt{(3 - Z_1)(3 + Z_1 + 2Z_2)}. \quad (45)$$

Under the specific LT-at-ISCO hypothesis,

$$R_{\text{in}} = R_{\text{ISCO}} = \rho_{\text{ISCO}}(q) \frac{G M_\infty}{c^2}. \quad (46)$$

Equations (42) and (46) contain two unknowns, q and M_∞ . The independent spectral R_{in} and withheld source-frame ν_{src} therefore determine a unique central exterior state without a CPTG-specific timing coefficient.

2.6.2 Primary and reverse closure

The primary test is ordered as

independent fitted $R_{\text{in}} \downarrow$ reveal withheld QPO $\downarrow \nu_{\text{obs}} \rightarrow \nu_{\text{src}} \downarrow$ solve Eqs. (42) and (46) $\downarrow$ derive $(M_\infty, q) \downarrow$ compare derived M_∞ with Eq. (4) $\downarrow$ apply fixed CPTG compact map.

A reverse cross-check starts instead from the separately reported mass median in Eq. (4), combines it with the withheld source-frame QPO, and predicts R_{ISCO} for comparison with Eq. (6). Because R_{in} and $M_{\bullet}$ arise from the same spectral posterior, the two directions are consistency checks, not statistically independent likelihood factors.

2.7 Numerical implementation and uncertainty propagation

All calculations use

$$G = 6.67430 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}, \quad c = 299792458 \text{ m s}^{-1}, \quad (47)$$

and

$$M_{\odot} = 1.98847 \times 10^{30} \text{ kg}. \quad (48)$$

The primary calculation uses the first-order comparison-coordinate sensitivity radius in Eq. (15) together with the source-frame frequency in Eq. (8). The central solution is obtained by one-dimensional root finding in q after eliminating M_{∞} with Eq. (46). The physical CPTG root of Eq. (21) is then solved on $1 \leq A_{\text{pol}} \leq 2$. Structural Mode and the angularly averaged compact readouts are evaluated numerically from the registered operators.

To propagate the published marginal uncertainties without pretending to possess the original **kerrSED** posterior chain, we also construct a summary-level Monte Carlo diagnostic with 300,000 draws. The asymmetric $\log_{10} R_{\text{in}}$ interval is represented by a two-sided Gaussian centered on 5.24 with lower and upper widths 0.05 and 0.13 dex, and each radius draw is transformed by Eq. (14); ν_{obs} is drawn from a Gaussian with standard deviation 0.38 mHz and transformed draw-by-draw by Eq. (7). This reconstruction reproduces the published marginal quantiles by construction but contains no unpublished covariance information. Its overlap fractions are therefore diagnostics, not formal posterior-predictive probabilities.

3 Results

The results are presented in the same order as the methodology: exterior closure first, independent spectral checks and branch discrimination second, and CPTG finite-state readouts only after the exterior state is fixed. The numerical values below therefore separate what is determined by the withheld spectral-timing construction from what follows downstream from the registered compact map.

3.1 Primary comparison-coordinate sensitivity, source-frame withheld closure

The primary spectral input is the first-order comparison-coordinate sensitivity radius

$$R_{\text{in}}^{\text{CPTG}} = 1.73130099 \times 10^5 \text{ km}, \quad (49)$$

and the withheld source-frame timing constraint is Eq. (8). Solving the Kerr LT-at-ISCO system gives

$$\begin{aligned} q &= 0.5005166, \\ M_\infty &= 2.77107 \times 10^4 M_\odot. \end{aligned} \tag{50}$$

with

$$\rho_{\text{ISCO}} = 4.230980, \quad R_{\text{ISCO}} = 4.230980 r_m. \tag{51}$$

The result is not obtained by inserting the separately reported spectral mass. That mass is revealed afterward. The timing-derived value in Eq. (50) lies inside the Guolo et al. 68% mass support after applying the same sub-percent coordinate-sensitivity scale,

$$2.034 \times 10^4 \lesssim M_\bullet^{\text{CPTG}}/M_\odot \lesssim 3.617 \times 10^4. \tag{52}$$

The corresponding unshifted published interval is 2.042×10^4 – $3.631 \times 10^4 M_\odot$; the distinction is negligible compared with the reported spectral uncertainty but is kept explicit.

Quantity	Central value	Role
Published R_{in}	1.7378×10^5 km	External spectral coordinate
CPTG sensitivity R_{in}	1.7313×10^5 km	Primary spectral input
Observed QPO frequency	11.88 ± 0.38 mHz	Withheld datum
Source-frame QPO frequency	12.5334 ± 0.4009 mHz	Source-frame timing coordinate
Exterior spin q	0.50052	Timing+radius solution
Exterior mass M_∞	$2.7711 \times 10^4 M_\odot$	Timing+radius solution
ρ_{ISCO}	4.23098	Retained Kerr exterior
CPTG sensitivity mass median	$2.4455 \times 10^4 M_\odot$	Independent SED-mass check

Table 2: Central first-order comparison-coordinate sensitivity, source-frame withheld-timing closure. The fitted disk radius and withheld timing datum determine the exterior state; the separately reported SED mass is retained as a check rather than reused as an input.

3.2 Published-summary uncertainty reconstruction

The summary-level Monte Carlo gives

$$q = 0.504^{+0.090}_{-0.040}, \tag{53}$$

and

$$M_\infty = 2.782^{+1.318}_{-0.388} \times 10^4 M_\odot, \tag{54}$$

where the displayed limits are the 16th and 84th percentiles of the reconstructed marginal diagnostic, not an original `kerrSED` posterior.

The corresponding ISCO coordinate is

$$\rho_{\text{ISCO}} = 4.217^{+0.154}_{-0.365}. \tag{55}$$

Approximately 73.5% of the reconstructed timing-plus-radius mass draws fall inside the first-order CPTG comparison-coordinate sensitivity interval derived from the Guolo et al. 68% mass support. Because the published R_{in} and $M_{\bullet}$ distributions originate from the same spectral fit and their full covariance is unavailable, this number is reported only as an overlap diagnostic.

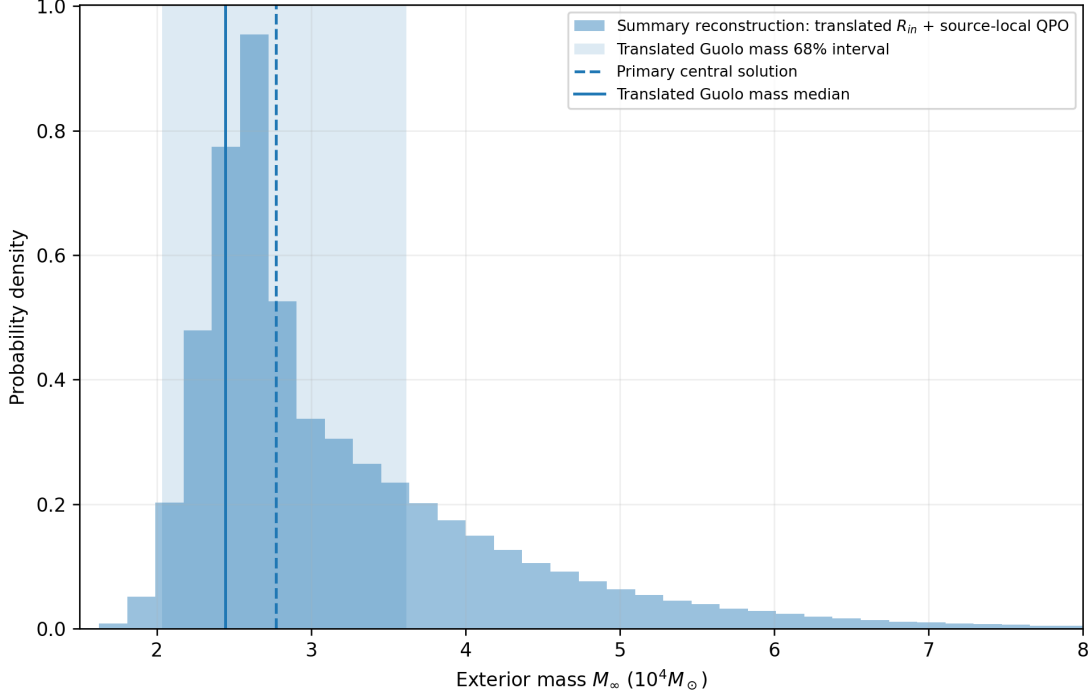


Figure 1: Published-summary closure diagnostic. The histogram is generated from the reported asymmetric R_{in} marginal and QPO measurement after transforming the timing frequency to the source frame. The shaded vertical region is the corresponding first-order CPTG comparison-coordinate sensitivity 68% spectral mass interval. The displayed mass axis is restricted to $1.5\text{--}8.0 \times 10^4 M_{\odot}$ for readability; all Monte Carlo draws are retained in the numerical calculation. This is not a reconstruction of the unpublished joint posterior.

3.3 Reverse mass-first closure

A useful reverse check fixes the Guolo et al. derived mass median,

$$M_{\infty} = 2.44553 \times 10^4 M_{\odot}, \quad (56)$$

and applies the same withheld source-frame frequency. The required exterior spin is

$$q = 0.470858, \quad (57)$$

which predicts

$$R_{\text{ISCO}} = 1.56950 \times 10^5 \text{ km}. \quad (58)$$

This falls inside the first-order CPTG comparison-coordinate sensitivity 68% spectral inner-radius interval of Eq. (16); it also lies inside the original published interval of Eq. (6). The primary and reverse constructions therefore close in both directions at the level supported by the published marginals.

3.4 Observed-frequency benchmark

For comparison, if the published observer-frame centroid 11.88 mHz is inserted directly without Eq. (7), the same R_{in} median yields

$$q = 0.485378, \quad M_{\infty} = 2.74321 \times 10^4 M_{\odot}. \quad (59)$$

This benchmark is close to the source-frame result because $z = 0.055$ is small, but the distinction is kept explicit. The observer-frame convention also reproduces the scale of the Zhang et al. published LT timing curve: at $q = 0.26$ and 0.36 it gives masses of approximately $9.62 \times 10^3 M_{\odot}$ and $1.59 \times 10^4 M_{\odot}$, respectively, close to their quoted joint range when combined with their same-epoch spectral contours.

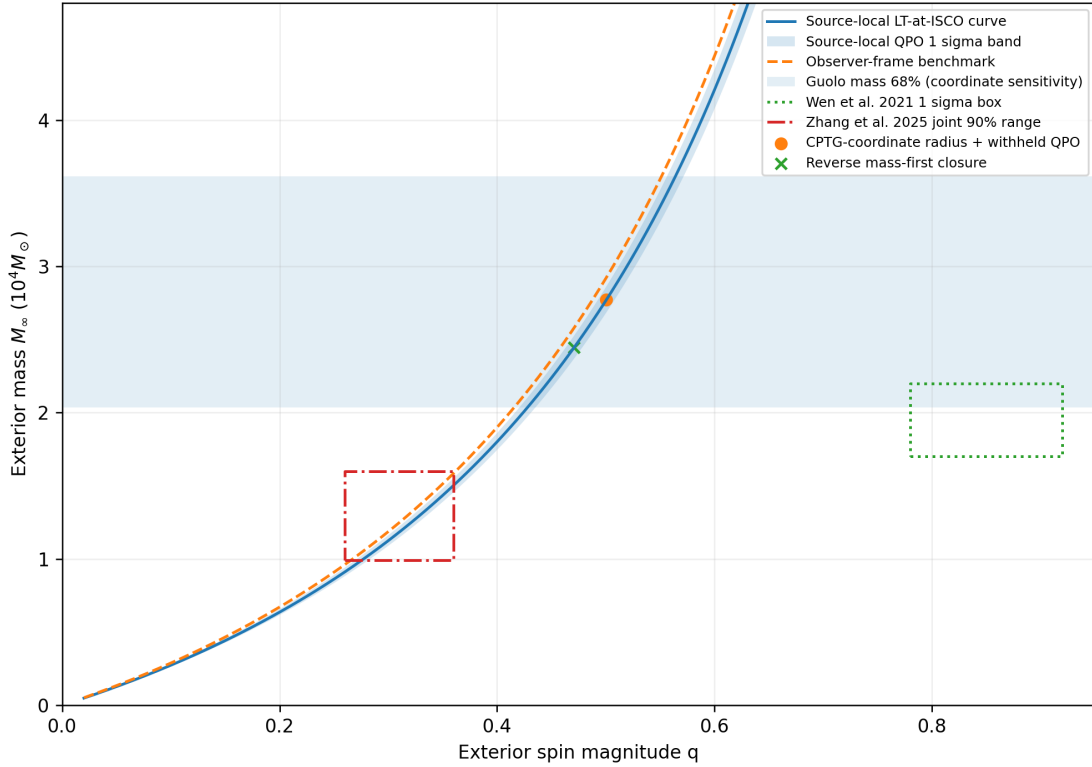


Figure 2: Mass–spin geometry of the timing constraint. The primary curve uses the source-frame QPO frequency. The observed-frequency curve is retained as a benchmark. Also shown are the first-order CPTG comparison-coordinate sensitivity interval derived from the Guolo et al. mass support, the earlier Wen et al. high-spin branch, and the Zhang et al. same-epoch joint range. Rectangles are visual summaries of published marginal ranges and should not be interpreted as joint posterior contours.

3.5 A different disk model as a branch-discrimination test

The earlier Wen et al. analysis used a relativistic slim-disk model and five X-ray epochs. For the $z = 0.055$ branch they reported

$$M_{\bullet} = 1.75^{+0.45}_{-0.05} \times 10^4 M_{\odot}, \quad q = 0.80^{+0.12}_{-0.02} \quad (60)$$

at 1σ [4]. This is a materially different disk construction from the later `kerrSED` analysis

and therefore provides a useful model-sensitivity comparison.

Using the Wen et al. central state in the same LT-at-ISCO timing operator predicts

$$\nu_{\text{LT}}^{\text{Wen,central}} = 76.17 \text{ mHz}, \quad (61)$$

which is

$$6.08 \times \nu_{\text{src}} \quad (62)$$

for the withheld source-frame QPO. At the same central mass, the source-frame timing signal would instead require

$$q = 0.39356. \quad (63)$$

Even the rectangular combination of the Wen et al. upper 1σ mass and lower 1σ spin,

$$M = 2.2 \times 10^4 M_{\odot}, \quad q = 0.78, \quad (64)$$

predicts

$$\nu_{\text{LT}} = 54.55 \text{ mHz} = 4.35 \times \nu_{\text{src}}. \quad (65)$$

This comparison does not falsify the Wen et al. spectral model as a whole. Their paper explicitly reports that the first two, highly super-Eddington epochs drive the high-spin combined solution and that separate early- and late-epoch spin constraints are inconsistent by more than 3σ under the fiducial hardening treatment [4]. The withheld timing signal therefore performs a useful role: conditional on LT-at-ISCO, it disfavors the high-spin five-epoch branch and selects the more moderate-spin region supported by the later independent SED construction.

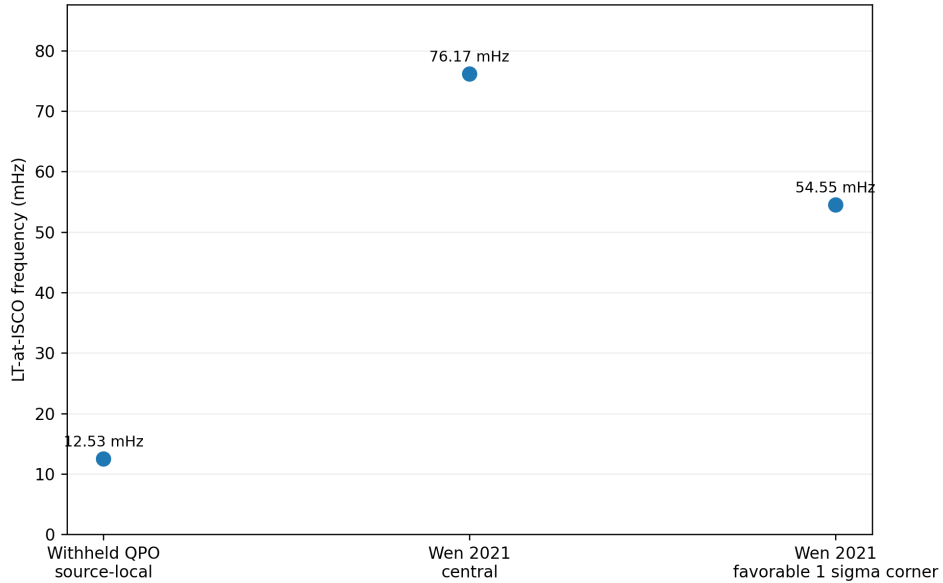


Figure 3: Timing discrimination among spectral branches under the common LT-at-ISCO hypothesis. The Wen et al. central high-spin branch and its favorable rectangular 1σ corner predict substantially higher source-frame LT frequencies than the withheld QPO.

3.6 CPTG finite-state decomposition

Only after the exterior state in Eq. (50) is fixed do we apply the CPTG compact relation. The physical root of Eq. (21) is

$$A_{\text{pol}} = 1.364108. \quad (66)$$

The corresponding source/geometric decomposition is

$$M_c = 2.03142 \times 10^4 M_{\odot}. \quad (67)$$

and

$$M_{\text{geom}} = 7.39656 \times 10^3 M_{\odot}. \quad (68)$$

Thus

$$\frac{M_{\text{geom}}}{M_{\infty}} = 0.26692. \quad (69)$$

or 26.69% of the exterior gravitational mass-equivalent in the registered handoff. The compact-source spin coordinate is

$$\hat{a}_c = 0.93136. \quad (70)$$

The summary-level uncertainty reconstruction gives the diagnostic percentiles shown in Table 3.

Quantity	16th percentile	Median	84th percentile
q	0.4645	0.5041	0.5944
$M_{\infty}/M_{\odot}$	2.394×10^4	2.782×10^4	4.100×10^4
A_{pol}	1.2723	1.3603	1.4035
$M_c/M_{\odot}$	1.707×10^4	2.045×10^4	3.222×10^4
$M_{\text{geom}}/M_{\odot}$	6.860×10^3	7.431×10^3	8.784×10^3
$M_{\text{geom}}/M_{\infty}$	0.2140	0.2649	0.2875
$\hat{a}_c$	0.9150	0.9328	0.9622

Table 3: Summary-level uncertainty propagation. The intervals reconstruct only the published one-dimensional R_{in} marginal and QPO error and are not substitutes for the unavailable original joint SED posterior.

3.7 Where the 85-second behavior occurs

For the central source-frame solution,

$$r_m = 4.09196 \times 10^4 \text{ km}. \quad (71)$$

The compact Structural Mode is

$$N_{\text{BH}} = 1.64421, \quad (72)$$

with

$$x_N = 0.529705, \quad r_N = 2.16753 \times 10^4 \text{ km.} \quad (73)$$

The inner trapping root is

$$x_- = 0.679801, \quad r_-^{\text{CPTG}} = 2.78172 \times 10^4 \text{ km.} \quad (74)$$

The retained Kerr outer horizon is

$$r_+^{\text{Kerr}} = 7.63448 \times 10^4 \text{ km,} \quad (75)$$

whereas the withheld-timing ISCO is

$$R_{\text{ISCO}} = 1.73130 \times 10^5 \text{ km.} \quad (76)$$

The physical ordering is therefore

$$r_N < r_-^{\text{CPTG}} < r_m < r_+^{\text{Kerr}} < R_{\text{ISCO}}. \quad (77)$$

In normalized form,

$$0.52971 < 0.67980 < 1 < 1.86572 < 4.23098. \quad (78)$$

This result resolves an important interpretive question. Under the tested LT-at-ISCO hypothesis, the 85-s behavior is not generated at the Structural-Mode shell, the inner trapping surface, or the finite-to-Kerr matching boundary. It occurs well outside the finite CPTG interior, in the retained Kerr exterior.

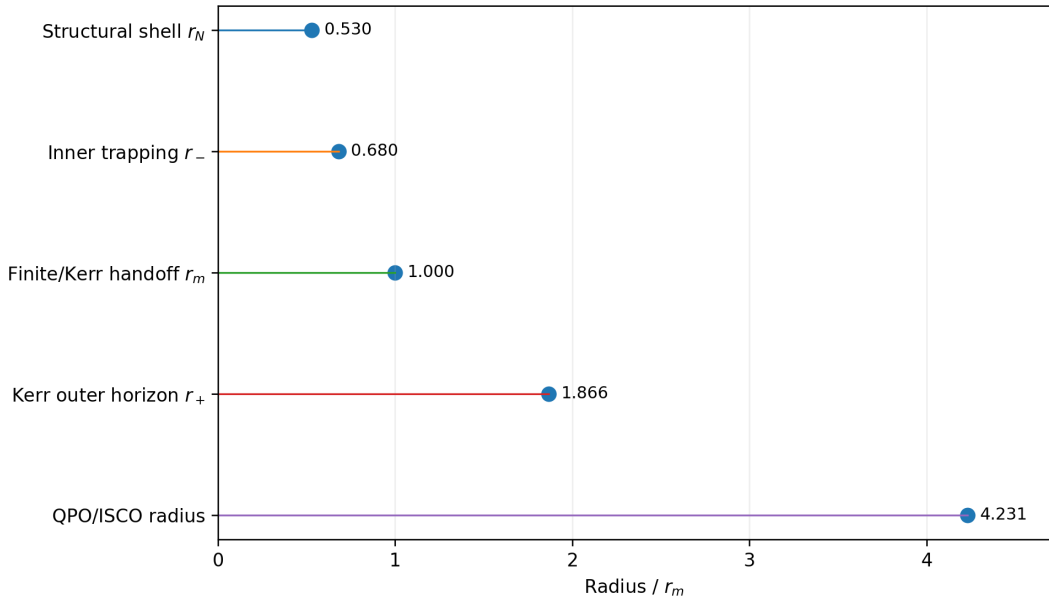


Figure 4: Normalized radial hierarchy for the central source-frame solution. The QPO/ISCO scale lies well outside the finite CPTG interior and the ordinary Kerr outer horizon.

The downstream organization-shell clock is

$$T_{24,N} = 16.7827 \text{ h} \quad (79)$$

per 24 source-frame asymptotic exterior Kerr hours. The determinant-averaged Structural-Mode frame-dragging rate is

$$\bar{\omega}_N = 3.51926 \text{ rad s}^{-1}. \quad (80)$$

For scale comparison only, $2\pi/\bar{\omega}_N = 1.785 \text{ s}$. This is a geometric frame-dragging timescale rather than a material rotation period and is not identified with the QPO. The compact theory defines it as a zero-angular-momentum-observer (ZAMO) readout [9, 10].

4 Discussion

The withheld construction is useful because it separates an observational exterior test from the later CPTG interpretation of that exterior state. This section first states what the closure establishes, then examines the disk-model discrimination, the finite-state consequences, and the specific observations that could strengthen or overturn the interpretation.

4.1 What the withheld closure establishes

The central result is an exterior consistency test. The timing signal is kept outside the compact construction until an independent disk scale has been specified, so the question is whether the later QPO selects an exterior mass–spin state consistent with the independently derived spectral mass. CPTG enters only after that observational state has closed.

Under the LT-at-ISCO identification, the answer is yes. The radius-first construction gives $M_\infty = 2.771 \times 10^4 M_\odot$ and $q = 0.5005$, while the reverse mass-first construction predicts an ISCO radius inside the independent spectral-radius interval. The small source-frame timing and luminosity-coordinate corrections do not create the closure. Removing the first-order CPTG distance sensitivity translation shifts the mass by less than half a percent; retaining the observer-frame QPO rather than the source-frame frequency still leaves the solution in the same moderate-spin, few- $10^4 M_\odot$ region.

This ordering separates the observational statement

$$(R_{\text{in}}, \nu_{\text{QPO}}) \longrightarrow (M_\infty, q) \quad (81)$$

from the CPTG statement

$$(M_\infty, q) \longrightarrow (M_c, M_{\text{geom}}, \hat{a}_c, N_{\text{BH}}, r_N, r_-). \quad (82)$$

No downstream CPTG quantity is allowed to flow backward into the exterior timing solve.

4.2 The LT-at-ISCO identification is a test hypothesis

The strongest limitation is physical rather than numerical. A QPO centroid is an observation; identifying it with a particular characteristic frequency is a model choice. Zhang et al. compare several Kerr characteristic frequencies at the ISCO and find that the LT-precession branch is the one that overlaps their same-epoch continuum constraints [1]. That makes LT at the ISCO a motivated hypothesis for 3XMM J2150, but not a theorem.

TDE disks can precess as extended flows, tear into independently precessing annuli, or produce quasi-periodic variability through mechanisms that are not reducible to a test particle exactly at the ISCO. A separate TDE, AT2020ocn, has been interpreted successfully with Lense–Thirring precession, yet alternative instability mechanisms remain possible [5]. More recently, Mummery and Quataert show that thick TDE disks capable of solid-body precession can undergo strong period evolution as the flow spreads, so a long-lived narrow-frequency QPO is not a generic prediction of global TDE-disk precession [6].

Every mass–spin number derived in this paper is therefore conditional on the local LT-at-ISCO identification. If a future physical model establishes that the 85-s feature corresponds instead to an extended-flow precession frequency, a disk-tearing mode, or another oscillation mechanism, the timing operator must be replaced and the exterior closure recomputed. The CPTG mass–spin relation itself would not be altered by that change; only the observational map into (M_∞, q) would change.

4.3 Why the second disk model strengthens the result

A single spectral model can conceal a systematic error by producing a narrow posterior. The comparison with Wen et al. therefore matters even though it does not reproduce the Guolo et al. likelihood. The two constructions make materially different assumptions and prefer different spin regimes. When the same withheld QPO operator is applied to both, the high-spin Wen branch predicts an LT-at-ISCO frequency several times larger than observed, while the later multi-epoch SED solution admits a moderate-spin closure.

This is a useful negative result. It shows that the QPO is not so weak a constraint that every published disk solution can be made compatible with it. Under the common LT-at-ISCO hypothesis, the timing signal selects among spectral branches. In that sense the QPO functions as an independent discriminator of accretion-model branches rather than merely as a datum that confirms a preselected spectral solution. The paper therefore benefits from model diversity rather than requiring every disk analysis to agree.

At the same time, the comparison must not be overstated. The Wen and Guolo fits use different epochs, disk physics, spectral-hardening treatments, and parameterizations. The QPO disagreement therefore diagnoses a branch inconsistency under a common timing hypothesis; it does not identify which individual spectral assumption is responsible.

4.4 What CPTG adds

Once the exterior state is fixed, the registered CPTG compact law makes a statement that ordinary Kerr exterior timing does not: the localized physical source need not equal the exterior gravitational mass parameter. For the central 3XMM state,

$$M_\infty = 2.7711 \times 10^4 M_\odot \quad (83)$$

is decomposed as

$$M_c = 2.0314 \times 10^4 M_\odot, \quad M_{\text{geom}} = 7.3966 \times 10^3 M_\odot. \quad (84)$$

The geometric share is therefore 26.69%. This value is not fitted to the QPO; it follows from Eq. (21) after the exterior spin is solved.

The same state fixes the internal organization and trapping hierarchy. Equation (77) shows that the observed QPO scale lies far outside the finite interior. This is the regime in which CPTG is required to recover the ordinary Kerr exterior. A successful timing closure is therefore a consistency test of the finite-to-Kerr construction rather than a detection of a modified exterior orbit.

4.5 What the exterior timing test establishes

The exterior QPO measures the gravitational state seen by matter and radiation outside the compact object. In the registered CPTG realization, that exterior is characterized by M_∞ and q and is intentionally Kerr-like. The timing result therefore fixes the exterior state but does not, by itself, separate the localized physical-source contribution from the completed geometric response that together supply the conserved exterior gravitational charge.

The 3XMM test goes beyond a simple overlap check because a fixed CPTG compact map, registered before this object was analyzed, accepts the independently closed exterior state and returns a definite nonzero geometric mass-equivalent contribution without object-specific retuning. The observation therefore supplies a newly constrained exterior state on which the CPTG source–geometry split makes a quantitative prediction.

Direct empirical discrimination of that split requires a second observable sensitive to the localized source or to strong-field structure that is not degenerate with the conserved exterior charges. Candidate classes include perturbative strong-field observables, interior-sensitive compact-state readouts, or an independently established physical mass budget that does not itself infer mass from the exterior gravitational field.

4.6 Why the location of the QPO matters

The hierarchy

$$r_N < r_- < r_m < r_+^{\text{Kerr}} < R_{\text{ISCO}} \quad (85)$$

eliminates one tempting but incorrect interpretation: the QPO is not naturally identified with the Structural-Mode clock or with the determinant-averaged frame-dragging rate at r_N . The Structural-Mode shell is an internal post-solution address. Its clock and ZAMO angular rate are local geometric readouts of the finite state; they are not generic X-ray modulation periods.

This separation strengthens the test because CPTG is not allowed to “explain” the QPO by assigning it to whichever internal timescale happens to be numerically convenient. The theory must first reproduce the observed exterior state in the regime where it overlaps Kerr. Only then are the finite-state readouts evaluated.

4.7 Comparison with the same-epoch Zhang solution

Zhang et al. obtain a lower-mass, lower-spin joint region by combining the QPO with continuum fits to the QPO-detection epoch itself [1]. Our preferred solution is not expected to reproduce that same joint contour because the purpose of the present experiment is different. We deliberately replace the same-epoch continuum constraint with an independent multi-epoch spectral construction.

The contrast is scientifically informative:

$$\text{same-epoch continuum} + \text{QPO} \quad \text{versus} \quad \text{independent SED} + \text{withheld QPO}. \quad (86)$$

The first maximizes simultaneity but shares spectral and timing systematics within one epoch. The second sacrifices simultaneity in exchange for independence of the primary spectral construction. The black-hole mass and spin should be intrinsic and stable, while the accretion flow evolves. Guolo et al. explicitly enforce fixed R_{in} and spin across their fitted epochs for the accretion regime they analyze [2]. The disagreement among spectral branches therefore motivates a dedicated reanalysis rather than invalidating the withheld design.

4.8 Falsifiable follow-up tests

The present calculation creates several direct failure modes. Release of the original `kerrSED` posterior samples would allow the exact joint posterior to be pushed through the withheld timing operator; the present result would be weakened if the apparent one-dimensional overlap disappeared once the true R_{in} –spin–inclination covariance were restored. An independent reanalysis of the non-QPO XMM1 and Chandra data with another relativistic disk model should recover a radius/mass region compatible with the moderate-spin timing branch if the closure is robust to spectral implementation. A persistent requirement for $q \gtrsim 0.8$ would conflict with the local LT-at-ISCO interpretation of the 85-s signal.

A future re-detection of the same QPO also supplies a timing-stability test. If the signal is tied to the ISCO of a black hole with fixed M_∞ and q , its source-frame centroid should remain approximately stable apart from measurement uncertainty and modest mode

physics. Strong secular chirping would favor an extended or evolving precession geometry over the local ISCO identification.

Finally, any future observable that independently constrains the localized physical source near

$$M_c \simeq 2.03 \times 10^4 M_\odot \quad (87)$$

rather than merely re-measuring M_∞ would directly test the source–geometry decomposition. Conversely, a robust determination that the localized physical source itself must equal the full exterior mass would falsify this registered CPTG compact branch for 3XMM J2150.

4.9 Remaining limitations

The Guolo et al. analysis is currently available as an arXiv preprint rather than a final journal record [2]. Its joint posterior samples are not used here, so the Monte Carlo intervals in Table 3 are summary-level reconstructions only. The exact comparison cosmology is incompletely specified beyond $H_0 = 73$ in the manuscript; our low-redshift comparison-coordinate sensitivity translation therefore exposes an explicit decoder assumption rather than hiding it, and its sub-percent effect demonstrates that the result is insensitive to that choice.

The physical interpretation of the QPO is also conditional. The LT-at-ISCO operator is motivated by the original timing paper but does not exclude other TDE variability mechanisms. Finally, the registered CPTG rotating finite realization remains a conditional compact handoff and downstream Kerr-compatible readout rather than a uniquely derived general axisymmetric interior solution from the parent action [9]. None of these limitations changes the numerical closure reported here, but all of them delimit what can be claimed from it.

5 Conclusion

3XMM J215022.4–055108 provides a rare compact-object test in which a spectral construction and a timing signal can be separated observationally. Using the independently fitted disk inner radius as the primary spectral quantity and withholding the XMM2 QPO until afterward, the source-frame 12.5334 ± 0.4009 mHz timing signal closes, under the LT-at-ISCO hypothesis, at

$$q = 0.5005, \quad M_\infty = 2.771 \times 10^4 M_\odot \quad (88)$$

after the small first-order CPTG luminosity comparison-coordinate sensitivity correction. The separately reported spectral mass and the reverse mass–first radius check are both consistent with that state at the level supported by the published marginals.

The same timing operator discriminates among spectral branches. The earlier high-spin slim-disk solution predicts an LT-at-ISCO frequency several times larger than the observed

source-frame QPO, whereas the later independent multi-epoch SED construction admits the moderate-spin closure. The value of the 85-s signal is therefore not merely that it permits an exterior closure, but that under a common timing hypothesis it discriminates among otherwise viable spectral constructions.

CPTG enters only after the exterior state is fixed. The registered compact mass–spin relation gives

$$M_c = 2.031 \times 10^4 M_\odot, \quad M_{\text{geom}} = 7.397 \times 10^3 M_\odot, \quad (89)$$

with a geometric mass-equivalent share of 26.7% and compact-source spin coordinate $\hat{a}_c = 0.9314$. The solved radial hierarchy places the QPO/ISCO well outside the finite CPTG interior, so the observed timing behavior is correctly treated as a Kerr-overlap exterior clock rather than an internal CPTG oscillation.

The result is therefore a withheld exterior-timing closure followed by a finite CPTG compact-state readout. Its significance is that a fixed source–geometry map, registered independently of this QPO, survives the timing test and assigns a definite physical source mass and geometric contribution without object-specific retuning. Direct empirical discrimination of that source–geometry split requires an observable sensitive to the localized source or finite interior rather than only to the conserved exterior gravitational charge. Exact posterior propagation, independent spectral re-fitting, repeated timing observations, and such interior-sensitive observables provide clear routes to strengthen or falsify the interpretation.

Data, Code, and Evidence Availability

The XMM-Newton and Chandra observation identifiers used by the cited spectral and timing studies are publicly archived. Zhang et al. state that the XMM-Newton source data are available through HEASARC and that the timing analysis uses the public Stingray software [1]. Guolo et al. identify the XMM-Newton and Chandra datasets used for the independent 3XMM J2150 SED construction [2].

The numerical calculations in this paper use only the equations stated in the manuscript and the registered CPTG compact relations cited above. The complete publication evidence archive is

`CPTG_3XMM_J215022_WITHHELD_TIMING_EVIDENCE.zip`

The archive contains the exact analysis script, numerical output table, figure source products, observational-input registry, CPTG-authority registry, reproduction command, and numerical audit used for the results reported here. The original `kerrSED` joint posterior samples are not reconstructed or approximated as though they were available; the reported uncertainty propagation is explicitly labeled as a published-summary diagnostic.

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